Module 2: Intelligent Data Driven Applications on the Cloud

INTELLIGENT DATA DEVELOPMENT, JUNE 2026

HANDS ON SESSION

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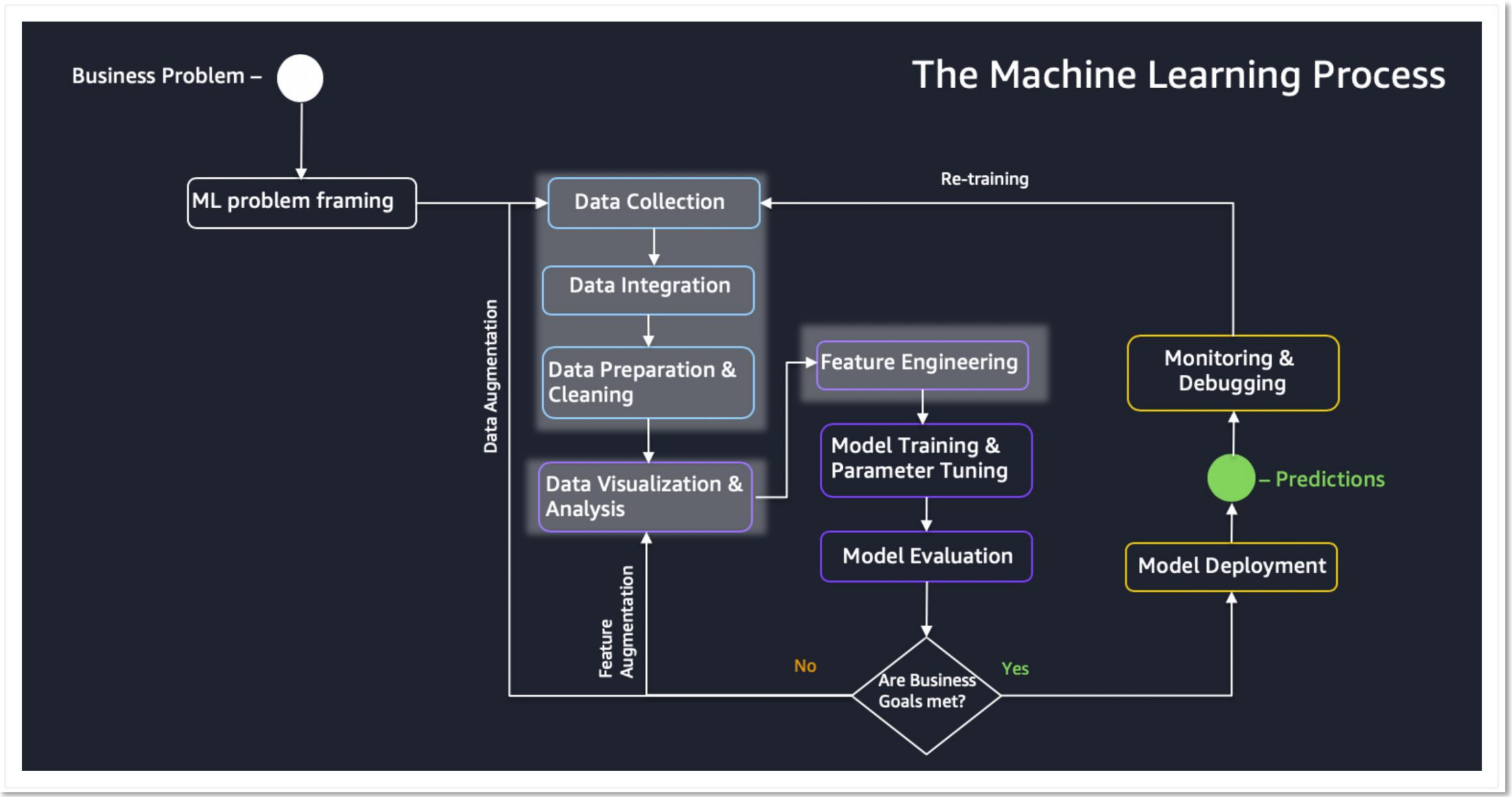
CLOUD MACHINE LEARNING



HANDS-ON CLOUD LAB: STEPS

1. ~~Connect to an AWS account using your IAM user – 10 mins~~
2. ~~Create an S3 bucket like idd-class-folder and load data.csv – 10 mins~~
3. ~~Delete the S3 bucket and empty it – 5 mins~~
4. Create a Lambda function by hand – 20 mins
5. Link the Lambda to S3 and give it proper permissions – 15 mins
6. Load data from s3 and save data in csv format into s3 – 20 mins
7. Launch all above with a CloudFormation template – 15 mins

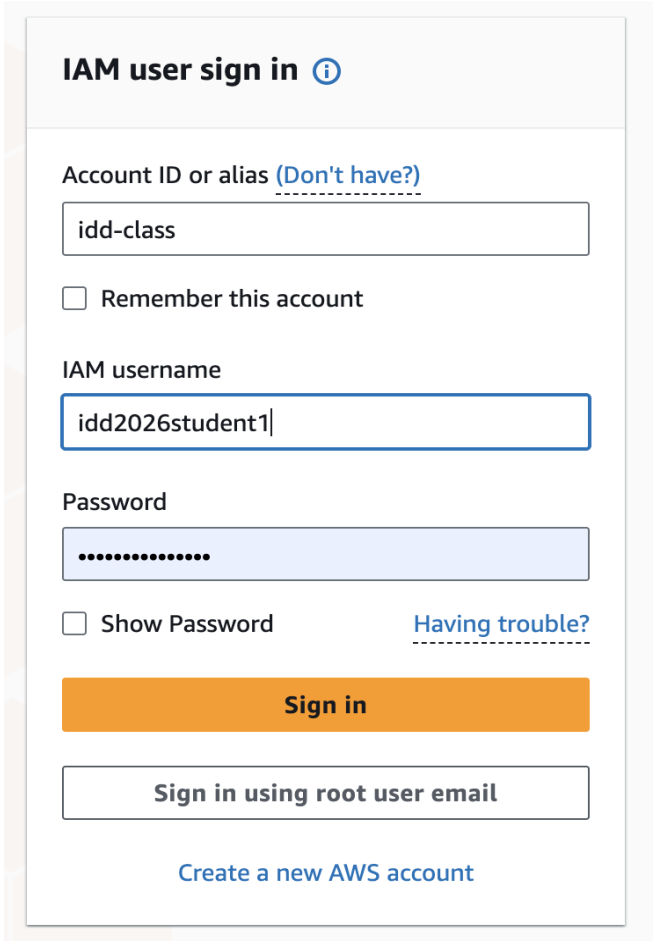
HANDS ON – STEP 1 - CONNECT TO AN AWS ACCOUNT USING AN IAM USER

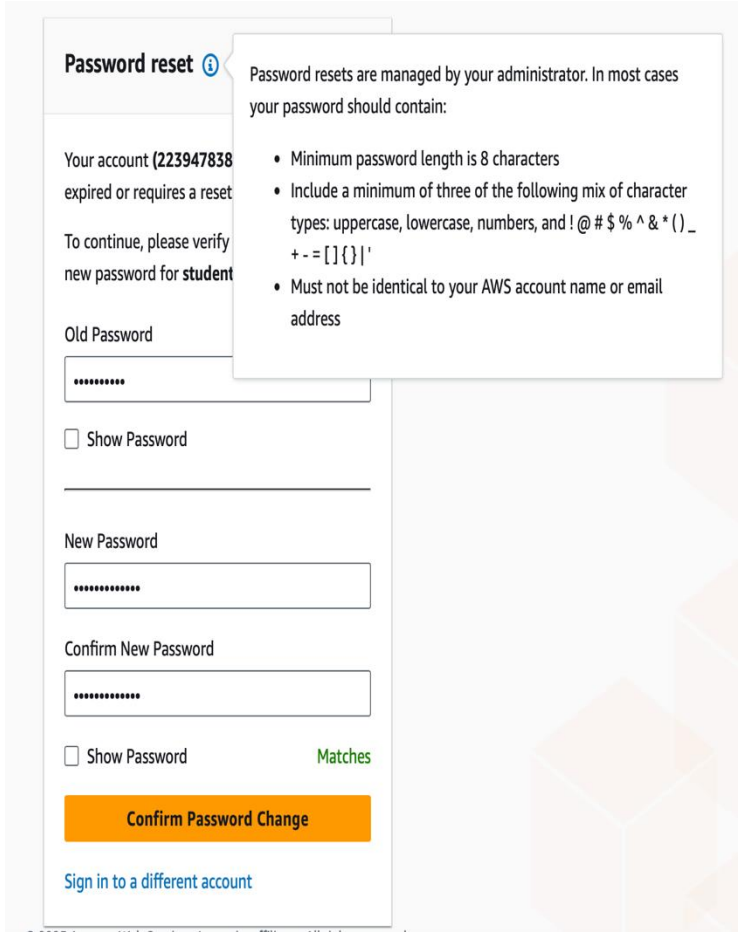
Step-by-step:

1. Go to the login link: **https://idd-class.signin.aws.amazon.com/console**
2. Log in using the credentials:
   - **Username**: ***name.surname/ name.middlename*** (or your assigned name)
   - **Password**: *ChangeMeNow_2024!*

3. On first login, set a **new password** you'll remember. A password reset screen will appear. You will need to add the old password provided and add your new one.

You're now inside your AWS account!





HANDS ON – STEP 2 – UPLOAD OUR DATA TO OUR S3 BUCKET

We'll add the train_data.csv to our own S3 buckets.

Steps:

1. Search for **S3** in the top search bar, click **S3**
2. Search for **<name>-<surname>-bucket-idd, i.e. laura-cozma-bucket-idd**
3. Upload data_train.csv

HANDS ON – STEP 3 - CREATE A LAMBDA FUNCTION BY HAND

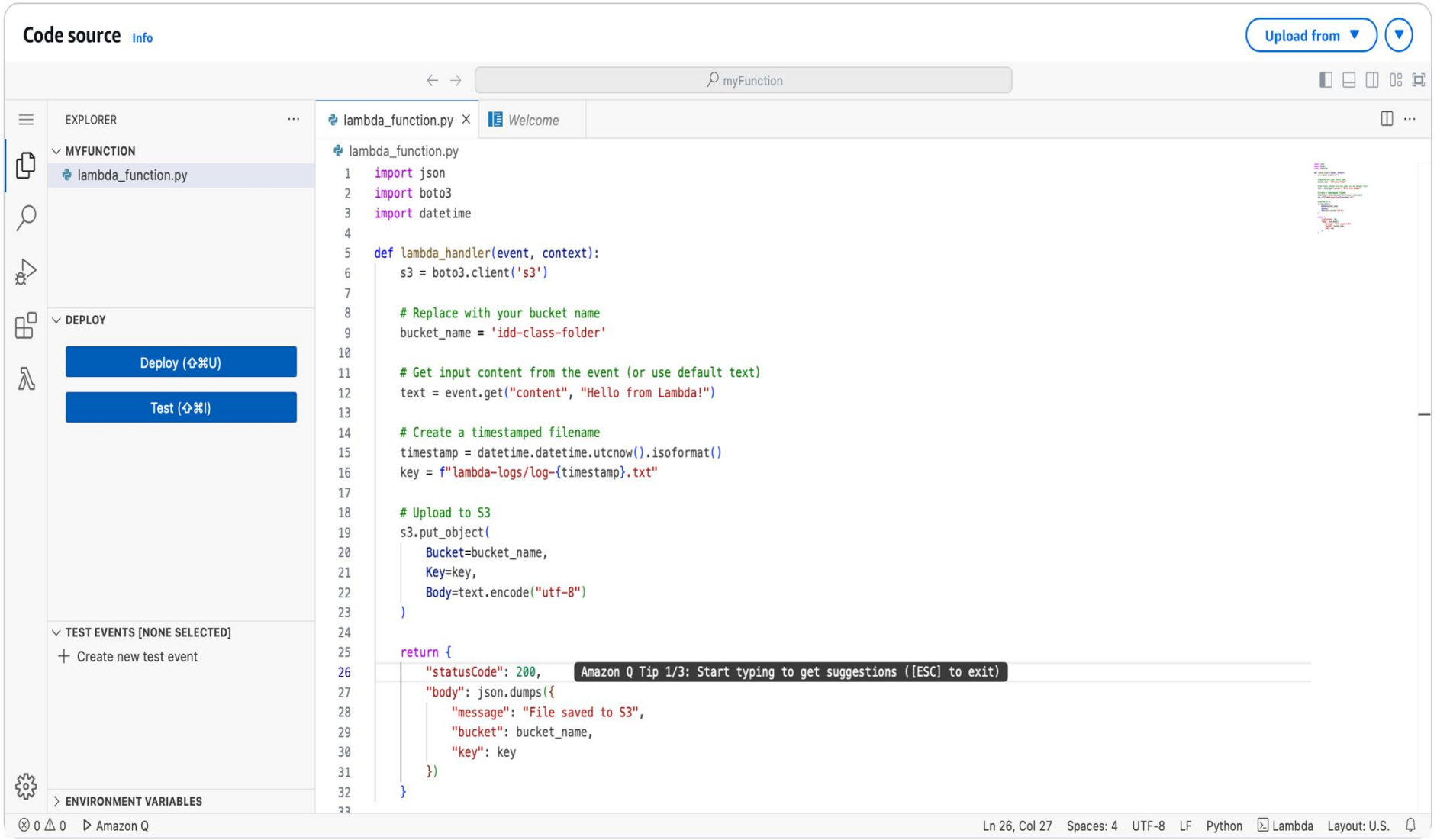
Step 1: Create an IAM Role for Lambda
1.Go to **IAM > Roles**
2.Click **Create role**
3.Choose **AWS service → Lambda**
4.Attach the following policies:
    1. AmazonS3FullAccess (or scoped-down access)
    2. AWSLambdaBasicExecutionRole
5.Name the role (e.g. LambdaS3Role) and create it.

Step 2: Create the Lambda Function
1.Go to **Lambda → Create function**
2.Choose:
    1. Name: ForecastingFunction
    2. Runtime: **Python 3.X**
    3. Execution role: Use existing role → select LambdaS3PandasRole
3.Create the function
4.Go to Configuration and set up time out more than 3 seconds.
5.Go to Layers and choose Pandas from the Layer's list.

HANDS ON – STEP 3 - CREATE A LAMBDA FUNCTION BY HAND

Step 3: Edit Lambda Code
Click **Code** and paste the following:
Replace your-input-bucket-name and file keys with your actual S3 path.

Step 4: Test the Lambda
1.Click **Test** → Create a test event (can be empty if hardcoded paths are used)
2.Run the Lambda
3.Check your S3 bucket to confirm the new file is saved



HANDS ON – STEP 3 - CREATE A LAMBDA FUNCTION BY HAND

Step 5: Package pandas for Lambda - optional
Lambda has a size limit for inline editing, and pandas isn't included by default. You must **deploy with a layer** or **zip upload**.
Option A: Use a Prebuilt Lambda Layer (Recommended for Console Users)
Use a public pandas layer:
1.Visit AWS Lambda Layer Library → Find the **pandas layer ARN** for your region (e.g. arn:aws:lambda:us-east-1:770693421928:layer:Klayers-p39-pandas:1)
2.Save the ARN for later

Step 6: Add pandas Layer to Lambda - optional
1.Open the Lambda function → Click **Layers** → **Add a layer**
2.Choose **Specify an ARN**
3.Paste your pandas layer ARN and click **Add**

LAMBA CODE

```
import json
import boto3
import datetime
import pandas as pd
from io import StringIO

def lambda_handler(event, context):
    s3 = boto3.client('s3')

    # Replace with your actual bucket name
    bucket_name = ‘name-surname-bucket-idd'
    input_key = ‘train_data.csv'

    try:
        # Read CSV from S3
        response = s3.get_object(Bucket=bucket_name, Key=input_key)
        content = response['Body'].read().decode('utf-8')
        df = pd.read_csv(StringIO(content))

        # Do any transformation here (this is just a placeholder)
        df_processed = df.copy()

        # Prepare to write back to S3
        timestamp = datetime.datetime.utcnow().isoformat()
        output_key = f"processed/output_{timestamp}.csv"
        csv_buffer = StringIO()
        df_processed.to_csv(csv_buffer, index=False)

        # Upload to S3
        s3.put_object(
            Bucket=bucket_name,
            Key=output_key,
            Body=csv_buffer.getvalue().encode('utf-8')
        )

        return {
            "statusCode": 200,
            "body": json.dumps({
                "message": "CSV processed and saved to S3",
                "bucket": bucket_name,
                "output_key": output_key
            })
        }

    except Exception as e:
        return {
            "statusCode": 500,
            "body": json.dumps({
                "error": str(e)
            })
        }
```

Things Change. Know Why - Key Use Case Examples

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THANK YOU!